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Studierichting: Informatica
Oefeningenreeks 4A nr. 22.1

$$\int \operatorname{Bgsin} x dx$$

$$\left\{ \begin{array}{l} u = \operatorname{Bgsin} x \\ dv = dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \dfrac{1}{\sqrt{1-x^2}} dx \\ v = x \end{array} \right.$$

$$\Rightarrow x \operatorname{Bgsin} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

$$u = 1 - x^2 \Rightarrow du = -2x dx$$

$$\Rightarrow x \operatorname{Bgsin} x + \frac{1}{2} \int \frac{1}{\sqrt{u}} du$$

$$= x \operatorname{Bgsin} x + \frac{1}{2} \int u^{-\frac{1}{2}} du = x \operatorname{Bgsin} x + \frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c$$

$$= x \operatorname{Bgsin} x + \sqrt{1-x^2} + c$$

References